

# Nexus State Warehouse

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# Problem

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Scientific data is fragmented:

- ▶ SAXS  $\rightarrow$  structure:  $p(r)$ ,  $R_g$ ,  $D_{max}$
- ▶ SPR  $\rightarrow$  kinetics:  $k_a$ ,  $k_d$
- ▶ Cryo-EM  $\rightarrow$  geometry: surface and curvature
- ▶ PCA  $\rightarrow$  thermodynamics: G-space and low-dimensional state maps

**Current failure mode:**

- ▶ no unified query layer
- ▶ no shared state representation
- ▶ no cross-domain inference object

# Vision

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Replace datasets with **states**

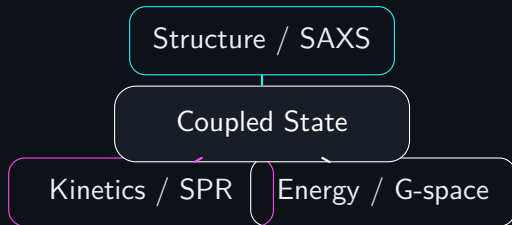
**State = minimal physically consistent description of a system**

Instead of querying experiments, Nexus queries biological and physical states that multiple experiments constrain together.

# Core Concept

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- ▶ Structure → SAXS
- ▶ Geometry → Cryo-EM
- ▶ Kinetics → SPR
- ▶ Energetics → PCA / G-space



# Architecture

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RAW DATA → FEATURES → STATES → WAREHOUSE →  
HYPOTHESES

The warehouse is the middle layer where heterogeneous experimental evidence becomes one queryable state object.

# Data Model

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- ▶ Dataset
- ▶ Feature
- ▶ State
- ▶ Model
- ▶ Hypothesis

These are not business tables. They are scientific abstractions for linking observables to admissible physical states.

# SQL Schema

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```
CREATE TABLE datasets (  
  dataset_id TEXT PRIMARY KEY,  
  method TEXT,  
  path TEXT,  
  temperature FLOAT,  
  ph FLOAT  
);
```

```
CREATE TABLE features (  
  feature_id TEXT PRIMARY KEY,  
  dataset_id TEXT,  
  name TEXT,  
  value FLOAT,  
  unit TEXT  
);
```

```
CREATE TABLE states (  
  state_id TEXT PRIMARY KEY,  
  system TEXT,  
  label TEXT,  
  confidence FLOAT
```

# Scientific Query Example

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**Question:** Does structural expansion correlate with stronger binding?

```
SELECT s.state_id
FROM states s
JOIN state_features sf1 ON s.state_id = sf1.state_id
JOIN features f1 ON sf1.feature_id = f1.feature_id
JOIN state_features sf2 ON s.state_id = sf2.state_id
JOIN features f2 ON sf2.feature_id = f2.feature_id
WHERE f1.name = 'Rg' AND f1.value > 5.0
      AND f2.name = 'kd' AND f2.value < 0.002;
```



# Hypothesis Layer

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## Example state:

- ▶  $R_g$  increases
- ▶  $k_d$  decreases

## Hypothesis:

- ▶ structural expansion stabilizes binding
- ▶ confidence is derived from multi-modal agreement

# Why This is Different

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## Classical data warehouse

- ▶ business metrics
- ▶ static tables
- ▶ dashboards

## Nexus warehouse

- ▶ physical states
- ▶ cross-domain coupling
- ▶ hypothesis generation

# Modern Stack

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## Core:

- ▶ DuckDB for local analytical execution
- ▶ Apache Arrow for columnar transport
- ▶ Python for orchestration and scientific execution

## Extensions:

- ▶ graph database for state relations
- ▶ feature store for ML integration
- ▶ versioned object storage for large experimental payloads

# Alternative Architectures

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## Graph-based:

- ▶ direct state-to-feature and state-to-state relations
- ▶ natural representation of coupled systems

## Lakehouse:

- ▶ versioned datasets
- ▶ time-travel queries
- ▶ reproducible warehouse snapshots

# Your Edge

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AI learns correlations

**Nexus learns physics**

# Conclusion

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**Nexus State Warehouse enables querying biological reality**